

DEVIKA YANAMALA

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PROFESSIONAL SUMMARY

Results-driven Data Analyst & Scientist with 2+ years of commercial experience building production ML/AI systems, BI dashboards, and data pipelines (Python, SQL, Power BI). Delivered £150K+ profit recovery and 30% efficiency gains. Skilled across the full data lifecycle: EDA, predictive modelling, NLP, deep learning, and stakeholder reporting. UK Graduate Visa eligible from Sept 2026.

TECHNICAL SKILLS

- AI & ML: XGBoost, Random Forest, Gradient Boosting, SMOTE, SHAP (Explainable AI), NLP, Deep Learning (LSTM, TensorFlow/PyTorch), Computer Vision, Time-Series Forecasting, RAG Pipelines, Scikit-Learn
- Analytics & BI: Power BI (Advanced DAX), Tableau, Advanced Excel (VBA, Power Query), SQL (CTEs, Window Functions), Pandas, NumPy, Matplotlib, Seaborn, EDA, Statistical Modelling
- Engineering & Cloud: Python, R, PostgreSQL, Snowflake, AWS (EC2, S3), React, Node.js, Git, Flask, Streamlit, APIs, GIS (GeoPandas, OSMnx)

PROFESSIONAL EXPERIENCE

Data Analyst (Studentship) | Newcastle University Library, UK

01/2026 – Present

- Engineered Python/SQL pipelines to process smartcard & occupancy datasets; deployed Power BI (DAX) dashboards to optimise library space allocation and reduce subscription costs.
- Collaborated with non-technical stakeholders to translate complex data patterns into actionable operational recommendations, improving data governance practices.

Lead Data Analyst & Developer | Rooted | Independent Project

04/2026 – Present

- Built a full-stack MVP (React, Node.js) with a location-based discovery engine integrating live geolocation APIs, serving 100+ cultural food communities across the UK.
- Designed an algorithmic fallback routing system handling data scarcity; applied Python/SQL to analyse user behaviour patterns for product personalisation.

Associate Data Scientist | BEPEC Solutions, Bangalore, India

05/2023 – 05/2025

- Developed and deployed production-grade ML models (classification, regression, NLP) in Python/AWS, improving operational efficiency by 30% and saving 10+ hrs/week via automated reporting.
- Led end-to-end client projects in Agile teams — from requirement gathering and EDA to model deployment and C-suite dashboard delivery.

Data Scientist Intern | BEPEC Solutions, Bangalore, India

01/2023 – 05/2023

- Performed EDA on large-scale datasets, validated predictive models and built visual reports; contributed to a 15% improvement in forecasting accuracy.

KEY PROJECTS

Next Best Action & Customer Churn Prediction (Banking) | Python · XGBoost · SHAP · SMOTE

- Built an end-to-end XGBoost churn pipeline with SHAP explainability and a Next Best Action recommendation engine mapping high-risk personas to retention products — aligned with UK banking regulatory standards.

Commercial Opportunities Analytics Dashboard (Retail) | Power BI · DAX · Python · Star Schema

- Engineered a Star Schema data model from 8,000+ transactional records; built a 20+ DAX-measure executive dashboard proving that capping discounts at 15% recovers £150K+ in annual profit.

Pharmaceutical Marketing Attribution & ROI Analysis | Python · ITS · Power BI · Regression

- Applied Interrupted Time Series (ITS) and regression attribution to identify a single marketing strategy driving \$1.3B+ in revenue; built an interactive Power BI Strategy Simulator with What-If DAX parameters.

MSc Dissertation: Automated OSM Topological Verification for ML-Based AADT Prediction | Python · OSMnx · GeoPandas · Random Forest · Gradient Boosting · Spatial CV

- Built a reproducible OSMnx pipeline to detect and correct 5 OSM topological error types (disconnected segments, grade-separated crossings, missing junction nodes, inconsistent one-way tags, road hierarchy errors) on 40,000+ node networks.
- Engineered graph-theoretic features (betweenness centrality, node degree, clustering coefficient) + Urban Atlas land-use variables; trained RF and Gradient Boosting AADT models with spatially blocked cross-validation, evaluating pre/post-correction performance via RMSE, MAE, MAPE and R².

EDUCATION

MSc Data Science & Artificial Intelligence | Newcastle University, UK | Expected 09/2026

Modules: Machine Learning, Deep Learning, Generative AI for Business, Advanced AI, Data Visualisation, Statistical Foundations, Image Processing

BEng (BTech) | IIIT RGUKT RK Valley, India | 2019–2023 | CGPA: 8.7/10

CERTIFICATIONS, ACHIEVEMENTS & VOLUNTEERING

- UK Work Culture Programme (Awarded Participant) | Newcastle University — Selected for a competitive 2-day programme; solved a live UK employer business challenge and presented actionable solutions to corporate stakeholders.
- Global Ambassador | WomenTech Network — Active in global initiatives promoting diversity, technical excellence, and leadership in AI & Data Science.
- Volunteer Module Representative | Newcastle University — Elected to represent the MSc cohort; collaborated directly with faculty to optimise curriculum delivery and student experience.